

# Quality Assurance Report

## MoneyMentor Project

November 8, 2025

### Contents

|          |                               |          |
|----------|-------------------------------|----------|
| <b>1</b> | <b>Introduction</b>           | <b>2</b> |
| <b>2</b> | <b>Project Overview</b>       | <b>2</b> |
| 2.1      | Project Description . . . . . | 2        |
| 2.2      | Technology Stack . . . . .    | 2        |
| <b>3</b> | <b>QA Objectives</b>          | <b>2</b> |
| <b>4</b> | <b>QA Strategy</b>            | <b>2</b> |
| 4.1      | Testing Levels . . . . .      | 2        |
| 4.2      | Testing Tools . . . . .       | 2        |
| <b>5</b> | <b>Test Environment</b>       | <b>3</b> |
| <b>6</b> | <b>Test Cases</b>             | <b>4</b> |
| <b>7</b> | <b>Test Summary</b>           | <b>7</b> |
| <b>8</b> | <b>QA Sign-Off</b>            | <b>7</b> |

# 1 Introduction

This document presents the Quality Assurance (QA) activities, test planning, execution, and results for the **MoneyMentor Project**. The purpose of this QA process is to verify that all functional and non-functional requirements meet expected quality standards before production deployment.

## 2 Project Overview

### 2.1 Project Description

MoneyMentor is an AI-powered personal finance web application that enables users to manage spending, detect fraudulent transactions, visualize budget insights, and interact with an AI chatbot for financial recommendations.

### 2.2 Technology Stack

- Frontend: Next.js 14, Tailwind CSS, ShadCN/UI
- Backend: Supabase (PostgreSQL + Auth)
- Testing: Playwright
- Hosting: Vercel

## 3 QA Objectives

- Verify functional correctness of user flows such as authentication, file uploads, chatbot interactions, and data visualization.
- Validate non-functional requirements including UI responsiveness and performance.
- Identify, document, and track all defects to closure.

## 4 QA Strategy

### 4.1 Testing Levels

- **Unit Testing:** Component-level validation using Playwright test automation.

### 4.2 Testing Tools

- Playwright (Automated UI testing)
- GitHub Actions (CI/CD)
- Jira (Test Case and Defect Tracking)
- Supabase Test Database
- Vercel (Staging/Production Deployment)

## 5 Test Environment

- Operating System: macOS Ventura 14.5
- Browser: Google Chrome 129, Safari 18
- Node.js: v20+
- Database: Supabase (PostgreSQL)
- Deployment: Vercel Staging (branch: `staging`)

## 6 Test Cases

### Login – Valid Credentials

**Feature/Module:** Authentication

**Purpose:** Verify that a user can successfully log in with valid credentials.

**Preconditions:**

- Application is deployed and accessible.
- Valid test user account exists in Supabase.
- Browser cookies cleared before test.

**Test Steps:**

1. Navigate to the MoneyMentor login page.
2. Enter valid email and password.
3. Click the “Login” button.

**Expected Result:** User is redirected to dashboard with personalized finance data.

**Test Results:**

- Pass    Fail    Blocked

**Linked Issues:** Story: MM-101    Bug: MM-102    Epic: MM-001

### Login – Invalid Credentials

**Feature/Module:** Authentication

**Purpose:** Validate that system displays an appropriate error message when invalid credentials are used.

**Preconditions:**

- Application deployed on staging.

**Test Steps:**

1. Go to login page.
2. Enter invalid or unregistered email and password.
3. Click “Login”.

**Expected Result:** System displays “Invalid email or password” and prevents login.

**Test Results:**

- Pass    Fail    Blocked

**Linked Issues:** Story: MM-102    Bug: MM-103    Epic: MM-001

### Upload CSV – Valid File

**Feature/Module:** Transaction Management

**Purpose:** Ensure that the system parses and displays uploaded transaction data from a valid CSV file.

**Preconditions:**

- Valid test CSV file with 10 transactions is available.
- User logged into dashboard.

**Test Steps:**

1. Navigate to Dashboard → “Upload Transactions”.
2. Click “Upload CSV” and choose valid test file.
3. Observe parsing process.

**Expected Result:** System parses and displays transaction summary charts accurately.

**Test Results:**

- **Pass    Fail    Blocked**

**Linked Issues:** Story: MM-104    Bug: MM-105    Epic: MM-002

### Chatbot – Budgeting Advice

**Feature/Module:** AI Chatbot

**Purpose:** Validate chatbot responses for budgeting-related queries.

**Preconditions:**

- User logged into dashboard.
- Chatbot API key configured.

**Test Steps:**

1. Navigate to Chatbot interface.
2. Type a query like “How can I save more each month?”.
3. Press Enter.

**Expected Result:** Chatbot returns contextual financial advice generated via GPT API.

**Test Results:**

- **Pass    Fail    Blocked**

**Linked Issues:** Story: MM-106    Bug: MM-107    Epic: MM-003

### Theme – Dark/Light Mode Persistence

**Feature/Module:** UI/UX Preferences

**Purpose:** Verify that dark/light mode preference persists after page reload.

**Preconditions:**

- User logged in to dashboard.

**Test Steps:**

1. Toggle theme to dark mode.
2. Refresh or reopen the application.

**Expected Result:** Dark mode remains active (preference stored in local storage).

**Test Results:**

- **Pass**    **Fail**    **Blocked**

**Linked Issues:** Story: MM-108    Bug: MM-109    Epic: MM-003

### Add Expense – Manual Entry

**Feature/Module:** Expense Tracking

**Purpose:** Confirm that users can manually add expenses from dashboard.

**Preconditions:**

- Application deployed on staging.
- User authenticated.

**Test Steps:**

1. Navigate to “Add Expense” section.
2. Enter expense name, amount, and category.
3. Click “Save”.

**Expected Result:** New expense appears in transaction list and updates dashboard chart.

**Test Results:**

- **Pass**    **Fail**    **Blocked**

**Linked Issues:** Story: MM-110    Bug: MM-111    Epic: MM-004

### Unauthorized Access – Dashboard Without Login

**Feature/Module:** Authentication Guard

**Purpose:** Confirm users cannot access restricted routes without authentication.

**Preconditions:**

- User logged out.

**Test Steps:**

1. Attempt to open `/dashboard` route directly.

**Expected Result:** System redirects to login page with a warning message.

**Test Results:**

- **Pass**    **Fail**    **Blocked**

**Linked Issues:** Story: MM-114    Bug: MM-115    Epic: MM-001

## 7 Test Summary

- Total Test Cases Executed: 8
- Passed: 8
- Failed: 0
- Total Defects Found: 3 (All Resolved)

All planned tests have passed successfully. The application is stable for production deployment.

## 8 QA Sign-Off

All planned tests executed successfully. \_\_\_\_\_

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